

Bulan 2: Advanced Python & Web Fundamentals

Minggu 1: Modules, Packages & Virtual Environments

Hari 1: Modules & Imports

Topik: import, from...import, creating modules, **name == "main"**

Proyek 1: MathToolkit

Custom math module

- Create math_utils.py module
- Functions: advanced_calc, statistics
- Import di main program
- Demonstrate module reusability

Proyek 2: StringToolkit

String utilities module

- text_processor.py dengan berbagai functions
- validators.py untuk validations
- Import & use dalam main app
- Teknologi: module creation, imports

Hari 2: Packages & init.py

Topik: Package structure, **init.py**, relative imports

Proyek: FinanceToolkit

Package untuk finance calculations

- Structure: finance_toolkit/**init.py**
- Module: currency.py (converter, formatter)
- Module: interest.py (simple, compound)
- Module: loan.py (calculator, amortization)
- Import functions dari package
- Demo aplikasi menggunakan package

Hari 3: Virtual Environments & pip

Topik: venv, pip install, requirements.txt, package management

Proyek 1: ProjectSetup

Setup environment best practices

- Create virtual environment
- Install packages (requests, python-dateutil)
- Generate requirements.txt
- Document setup process

Proyek 2: PackageExplorer

Explore installed packages

- List installed packages
- Check versions
- Test import berbagai packages
- Create project template structure

Hari 4: Working with External Libraries

Topik: requests library, API calls, JSON parsing

Proyek 1: WeatherApp

App cuaca dari API

- Call weather API (OpenWeatherMap/WeatherAPI)
- Parse JSON response
- Display current weather
- Format output user-friendly
- Teknologi: requests, JSON

Proyek 2: CurrencyConverter

Real-time currency converter

- Call exchange rate API
- Convert between currencies
- Cache rates (save to JSON)
- Handle API errors

Hari 5: datetime & time modules

Topik: datetime objects, timedelta, strftime, strptime, timezone

Proyek 1: TimeTracker

Pelacak waktu produktivitas

- Start/stop timer
- Calculate duration
- Log activities dengan timestamp
- Daily/weekly reports
- Save to JSON

Proyek 2: EventReminder

Pengingat event

- Add events dengan date & time
- Calculate days until event
- Check upcoming events (next 7 days)
- Timezone handling
- Teknologi: datetime, timedelta, JSON

Hari 6: Regular Expressions (Regex)

Topik: re module, patterns, search, match, findall, sub

Proyek 1: EmailValidator

Validasi email dengan regex

- Pattern untuk email validation
- Extract emails dari teks
- Validate format
- Clean & normalize emails

Proyek 2: DataExtractor

Extract data dari teks

- Extract phone numbers (berbagai format)
- Extract URLs
- Extract dates

- Extract prices/currency
- Teknologi: re module, patterns

Hari 7: Review & Mini Project

Mini Project: APIDataAggregator

Agregator data dari multiple APIs

- Fetch data dari 3+ public APIs (weather, quotes, news)
- Parse & combine data
- Cache responses (JSON)
- Error handling (timeout, invalid response)
- Schedule periodic updates
- Generate combined report
- Export to CSV & JSON
- Teknologi: requests, json, csv, datetime, error handling, modules

Minggu 2: HTML & Web Basics

Hari 8: HTML Basics

Topik: Tags, elements, attributes, document structure

Proyek 1: PersonalProfile.html

Halaman profil pribadi

- Semantic HTML (header, main, footer)
- Headings, paragraphs, lists
- Links & images
- Basic structure

Proyek 2: BlogPost.html

Halaman artikel blog

- Article structure
- Multiple sections
- Blockquotes
- Code blocks dengan pre tag
- Author info

Hari 9: HTML Forms & Tables

Topik: Forms, input types, labels, tables, semantic HTML

Proyek 1: ContactForm.html

Form kontak lengkap

- Text inputs (name, email, phone)
- Textarea untuk message
- Select dropdown (subject)
- Radio buttons, checkboxes
- Submit button
- Form validation attributes

Proyek 2: DataTable.html

Tabel data produk

- Table dengan header
- Multiple rows produk
- Colspan & rowspan
- Sortable headers preparation
- Caption & summary

Hari 10: CSS Basics (untuk Backend Developer)

Topik: Selectors, properties, box model, basic styling

Note: Fokus minimal CSS untuk understand template structure

Proyek 1: StyledProfile.html

Profile dengan basic styling

- Inline, internal, external CSS
- Colors, fonts, spacing
- Basic layout
- Hanya perlu tahu konsep, tidak perlu bagus

Proyek 2: FormStyling.html

Form dengan basic CSS

- Input styling

- Button styling
- Basic responsive
- Understand class & id selectors

Hari 11: HTTP Protocol Basics

Topik: HTTP methods (GET, POST, PUT, DELETE), status codes, headers

Proyek 1: HTTPExplorer

Explore HTTP dengan requests

- Send GET requests
- Send POST dengan data
- Check response status codes
- Inspect headers
- Handle different status codes
- Teknologi: requests library

Proyek 2: APITester

Test REST API endpoints

- Test CRUD operations ke dummy API
- GET, POST, PUT, DELETE requests
- Parse responses
- Error handling by status code
- Log all requests & responses

Hari 12: JSON & REST API Concepts

Topik: JSON structure, REST principles, API endpoints

Proyek 1: RESTConsumer

Consume REST API

- Fetch dari JSONPlaceholder API
- GET users, posts, comments
- Filter & search data
- Nested data handling
- Teknologi: requests, JSON parsing

Proyek 2: MockAPIServer

Simulasi REST API sederhana

- Read data dari JSON file
- Simulate API endpoints (manual routing)
- Return JSON responses
- Basic CRUD simulation
- Teknologi: JSON, file I/O, functions

Hari 13: SQL Basics - Part 1

Topik: Databases intro, SQL syntax, SELECT, WHERE, ORDER BY

Proyek 1: SQLiteExplorer

Explore database dengan Python

- Connect ke SQLite database
- Execute SELECT queries
- Filter dengan WHERE
- Sort hasil
- Display results formatted
- Teknologi: sqlite3 module

Proyek 2: ProductCatalog.db

Database katalog produk

- Create products table
- INSERT sample data (20+ products)
- Query by category
- Search by name
- Price range filtering
- Teknologi: sqlite3, SQL queries

Hari 14: SQL Basics - Part 2

Topik: INSERT, UPDATE, DELETE, CREATE TABLE, data types

Proyek: InventoryDatabase

Database inventory lengkap

- CREATE TABLE untuk products, categories
- INSERT multiple records
- UPDATE stock quantities
- DELETE discontinued items
- Query relationships
- Generate reports dari SQL
- Export to CSV
- Teknologi: sqlite3, SQL CRUD, CSV

Minggu 3: Advanced SQL & Database Design

Hari 15: SQL Joins

Topik: INNER JOIN, LEFT JOIN, RIGHT JOIN, relationships

Proyek: OrderManagementDB

Database sistem order

- Tables: customers, orders, order_items, products
- Create related tables
- INSERT data dengan relationships
- JOIN queries (customer orders)
- Aggregate functions (SUM, COUNT, AVG)
- Complex queries (customer dengan total spending)
- Teknologi: sqlite3, JOIN operations

Hari 16: SQL Aggregations & Grouping

Topik: GROUP BY, HAVING, COUNT, SUM, AVG, MIN, MAX

Proyek: SalesAnalyticsDB

Database analytics penjualan

- Sales transactions table
- GROUP BY category, date, region
- Calculate totals & averages
- Top selling products
- Monthly sales trends

- HAVING untuk filter groups
- Teknologi: sqlite3, aggregate functions

Hari 17: Database Design & Normalization

Topik: Primary keys, foreign keys, normalization, ER diagrams

Proyek: SchoolManagementDB

Design database sekolah

- Tables: students, teachers, classes, enrollments, grades
- Design schema (normalize to 3NF)
- Define relationships
- Create tables dengan constraints
- INSERT sample data
- Complex queries (student grades per class)
- Teknologi: database design, sqlite3

Hari 18: Indexes & Query Optimization

Topik: CREATE INDEX, query performance, EXPLAIN

Proyek: PerformanceTestDB

Test query performance

- Large dataset (10,000+ records)
- Queries tanpa index
- Add indexes
- Compare performance
- EXPLAIN query plans
- Optimize slow queries
- Teknologi: sqlite3, indexing

Hari 19: Transactions & Data Integrity

Topik: BEGIN, COMMIT, ROLLBACK, ACID properties

Proyek: BankingTransactionDB

Simulasi transaksi banking

- Accounts table

- Transfer function (debit & credit)
- Use transactions (ROLLBACK on error)
- Ensure data integrity
- Transaction logs
- Teknologi: sqlite3, transactions

Hari 20: Database ORM Introduction

Topik: ORM concepts, SQLAlchemy basics

Proyek 1: ORMBasics

Intro to SQLAlchemy

- Install SQLAlchemy
- Define models (classes)
- Create tables from models
- Basic CRUD dengan ORM
- Query dengan ORM

Proyek 2: BlogORM

Blog dengan SQLAlchemy

- Models: User, Post, Comment
- Relationships
- Create, read, update, delete
- Query dengan filters
- Teknologi: SQLAlchemy ORM

Hari 21: Review & Mini Project

Mini Project: EcommerceDatabase

Database e-commerce lengkap

Tables:

- users (id, email, password_hash, created_at)
- categories (id, name, description)
- products (id, name, price, stock, category_id)
- orders (id, user_id, total, status, created_at)

- order_items (id, order_id, product_id, quantity, price)
- reviews (id, product_id, user_id, rating, comment)

Features:

- Complete schema dengan relationships
- Sample data (50+ products, 20+ users, 100+ orders)
- Complex queries (top selling, purchase history, ratings, revenue)
- ORM models dengan SQLAlchemy
- CRUD functions untuk semua tables
- Transaction handling untuk orders
- Export reports to CSV
- Teknologi: SQLite, SQLAlchemy, SQL queries, transactions, CSV

Minggu 4: Python Advanced Topics & Preparation for Django

Hari 22: Decorators

Topik: Function decorators, @wraps, decorator with arguments

Proyek 1: TimingDecorator

Measure function execution time

- Create @timer decorator
- Apply ke berbagai functions
- Log execution times
- Compare performance

Proyek 2: AuthDecorator

Simulasi authorization decorator

- @require_auth decorator
- Check user permissions
- Log access attempts
- Prevent unauthorized access
- Teknologi: decorators, functools

Hari 23: Context Managers

Topik: with statement, **enter**, **exit**, contextlib

Proyek 1: FileManager

Custom file context manager

- Auto open/close files
- Error handling
- Logging
- Use with statement

Proyek 2: DatabaseConnection

DB connection context manager

- Auto connect/disconnect
- Transaction management
- Error rollback
- Resource cleanup
- Teknologi: context managers, sqlite3

Hari 24: Multithreading Basics

Topik: threading module, Thread class, daemon threads

Proyek 1: ParallelDownloader

Download multiple files parallel

- Download 5 files simultaneously
- Progress tracking per thread
- Thread synchronization
- Teknologi: threading, requests

Proyek 2: BackgroundTasks

Background task processor

- Main program runs
- Background threads untuk periodic tasks
- Daemon threads
- Thread coordination

- Teknologi: threading, time

Hari 25: Working with Environment Variables

Topik: os.environ, python-dotenv, configuration management

Proyek 1: ConfigManager

Centralized config management

- Load dari .env file
- Environment-specific configs (dev, prod)
- Secret management
- Config validation
- Teknologi: python-dotenv, os

Proyek 2: SecureAPIClient

API client dengan env vars

- API keys dari environment
- Database credentials dari .env
- Different configs per environment
- Never hardcode secrets
- Teknologi: python-dotenv, requests

Hari 26: Logging

Topik: logging module, levels, handlers, formatters

Proyek 1: ApplicationLogger

Comprehensive logging system

- Different log levels (DEBUG, INFO, WARNING, ERROR)
- File handler & console handler
- Formatted logs dengan timestamps
- Rotate log files
- Teknologi: logging module

Proyek 2: ErrorTracker

Error tracking & monitoring

- Log semua errors

- Email alerts untuk critical errors (simulate)
- Error statistics
- Log analysis tools
- Teknologi: logging, smtplib

Hari 27: Testing with unittest

Topik: unittest module, test cases, assertions, setUp/tearDown

Proyek 1: CalculatorTests

Test calculator functions

- Test basic operations
- Test edge cases
- Test error handling
- Coverage report
- Teknologi: unittest

Proyek 2: APIClientTests

Test API client

- Mock API responses
- Test success & error cases
- Test data validation
- Integration tests
- Teknologi: unittest, unittest.mock

Hari 28: Review & Final Project Bulan 2

Final Project: TaskManagementAPI

RESTful task manager (console-based simulation)

Features:

- User registration & authentication (hash passwords)
- CRUD tasks (title, description, priority, due_date, status)
- Categories & tags
- Search & filter tasks
- SQLite database dengan SQLAlchemy

- Transaction handling
- Logging semua operations
- Unit tests untuk core functions
- Environment-based config (.env)
- Export tasks to JSON/CSV
- Import tasks dari JSON/CSV
- Statistics & reports
- Background reminder system (threading)

Project Structure:

```
task_manager/
  config/
    settings.py
  models/
    user.py, task.py, category.py
  services/
    auth_service.py, task_service.py
  utils/
    logger.py, validators.py
  tests/
    test_auth.py, test_tasks.py
  main.py
  requirements.txt
  .env.example
```

Teknologi: SQLAlchemy, sqlite3, logging, unittest, threading, python-dotenv, decorators, context managers, error handling, JSON, CSV

Progress Check Bulan 2: ✅ Modules & packages ✅ Virtual environments ✅ External libraries (requests)
 ✅ HTML basics (minimal untuk backend) ✅ HTTP & REST concepts ✅ SQL (queries, joins, optimization)
 ✅ Database design & normalization ✅ SQLAlchemy ORM ✅ Advanced Python (decorators, context managers) ✅ Testing, logging, config management

Total Proyek: 55+ proyek & 2 major projects **Siap lanjut:** Bulan 3 - Django Fundamentals